

VINAY KUMAR REDDY SIRIGIREDDY

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PROFESSIONAL SUMMARY

Aerospace systems engineer with an M.S. from Wright State University and graduate research focused on eVTOL conceptual sizing, MBSE-driven architecture, and computational fluid dynamics — including a physics-validated browser-based sizing engine deployed at [evtolsizer.live](#). Built and closed MATLAB sizing loops covering actuator-disk hover power, cruise power loading, battery energy capacity, and ISA-corrected noise estimation, benchmarked against published manufacturer data across multiple reference eVTOL configurations. Proficient in CATIA Magic SysML, OpenVSP, NASA CEA, and numerical methods for fluid dynamics, with hands-on experience translating mission requirements into quantifiable system parameters across propulsion, trajectory analysis, and CAD design domains.

TECHNICAL CORE COMPETENCIES

Aerospace Frameworks & Sizing: eVTOL Conceptual Sizing, Actuator Disk Theory, Power Loading Analysis, Disk Loading Optimization, Battery Energy Density Trade Studies, Class I/II Weight Estimation, Hover & Cruise Power Modeling, ISA Atmospheric Correction, Rotor Noise Estimation, Configuration Trade Studies

Computational & Numerical Methods: 1D Euler Equations (FVM/FDM), Advection-Diffusion & Burgers Equation, Upwind/Lax-Wendroff/Crank-Nicolson Schemes, CFL Stability Analysis, CFD (Cradle CFD), XFOIL, OpenVSP, HyperMesh, FEA, NASA CEA

Systems Engineering & MBSE: CATIA Magic SysML, MBSE, SysML, TPM Tracking, FMEA, Fault Tree Analysis, PLM, 3D Swym, Configuration Management

Design & Simulation Tools: SOLIDWORKS (CSWP/CSWA), CATIA V5, AutoCAD (2D & 3D), Fusion 360, Onshape, ANSYS, STK (Systems Tool Kit)

Programming & Development: MATLAB, Python (NumPy, SciPy, Pandas), JavaScript (React, Vite), HTML/CSS, MS Excel

PROFESSIONAL EXPERIENCE

Graduate Research Assistant — Wright State University, Fairborn, OH

Feb 2025 – Apr 2026

- Derived a physics-based preliminary sizing algorithm in MATLAB using actuator disk theory for hover power, a lift-to-drag ratio method for cruise power, and Peukert-corrected battery capacity estimation, converging MTOW predictions for three eVTOL reference configurations within $\pm 5\%$ of published manufacturer data.
- Translated the MATLAB sizing framework into a publicly deployed browser-based tool ([evtolsizer.live](#)) using React and Vite, restructuring ~1,200 lines of procedural MATLAB into modular JavaScript functions and enabling real-time parametric sweep studies across rotor count, disk loading, and energy density inputs.
- Parameterized CATIA Magic SysML block definition and internal block diagrams to track Technical Performance Measures — disk loading (lb/ft^2), hover figure of merit, and specific energy (Wh/kg) — across 10+ eVTOL design iterations, mapping propulsion-battery-flight control interface dependencies and identifying four single-point failure modes before the detailed design gate.
- Generated parametric eVTOL airframe geometries in OpenVSP for five candidate configurations and extracted aerodynamic coefficients (CL, CD, L/D) at cruise conditions, using these as direct inputs to close the drag-power-weight iteration loop and reduce sizing residuals by two full Newton iterations per configuration.
- Solved the 1D advection-diffusion and viscous Burgers equations using finite difference schemes (upwind, Lax-Wendroff, Crank-Nicolson), quantifying numerical diffusion and phase error as functions of CFL number and mesh refinement ratio across a structured 3-level grid refinement study.
- Authored a research manuscript submitted to the AIAA Journal of Aircraft documenting the integrated MBSE-driven sizing methodology, writing the numerical framework derivation, three-vehicle validation study, and results sections in full.

Advanced Rocket Propulsion Intern — MARS Exploration Pvt. Ltd., Chennai, TN, India *May 2022 – Aug 2022*

- Evaluated combustion performance of RP-1/LOX and AP/Al propellant systems using NASA CEA, computing specific impulse (Isp), characteristic velocity (c^*), and adiabatic flame temperature across O/F ratios from 1.0 to 3.5 to map the thermodynamic efficiency surface and locate the peak performance point.
- Characterized O/F ratio sensitivity for an AP/Al composite solid propellant formulation, identifying an optimal range of 1.5–2.0 that produced approximately 12% improvement in computed Isp relative to the nominal mixture ratio baseline used in the prior design cycle.
- Modeled ion thruster performance parameters — specific impulse, thrust efficiency, and propellant utilization fraction — from published test data, comparing electric propulsion trade space against chemical alternatives for a defined delta-V mission budget to support propulsion system down-selection.

- Specified CNC machining tolerances, surface finish requirements (R_a in μin), and material property inputs for high-pressure combustion chamber components, translating thermodynamic boundary conditions from CEA output into manufacturing drawing callouts for shop-floor fabrication.

Aerospace Systems Analyst (STK) Intern — SS Technologies, Bengaluru, Karnataka, India Nov 2021 – Feb 2022

- Modeled five satellite orbits in STK using two-body and J2-perturbed propagators, computing ground track coverage, revisit intervals, and eclipse durations across three orbital inclinations to evaluate mission feasibility against a defined minimum daily coverage requirement.
- Simulated aircraft trajectory profiles in STK to assess radar line-of-sight geometry, ground station contact windows, and link access durations, generating structured access data tables used to validate communication architecture assumptions in the preliminary design review package.
- Translated mission requirement documents into STK scenario parameters — orbit altitude, inclination, sensor field-of-regard half-angle, and minimum elevation mask — establishing direct traceability between mission objectives and simulation inputs for three separate mission concepts.
- Compiled analysis results and 3D mission visualizations into technical briefing packages, consolidating coverage gap findings and system interaction constraints into a format that reduced review preparation time on successive scenario runs.

2D & 3D CAD Design Intern — Verzeo, Bengaluru, Karnataka, India Aug 2021 – Oct 2021

- Modeled multi-component mechanical assemblies in SOLIDWORKS using parametric feature-based design, applying mates and constraints to validate kinematic behavior and detect geometric interference conditions across all assembly configurations before physical prototyping.
- Produced detailed engineering drawings with GD&T callouts per ASME Y14.5, specifying flatness, perpendicularity, and positional tolerances on critical mating surfaces to define unambiguous manufacturing requirements for machined part families.
- Performed worst-case tolerance stack-up analysis on two-part slip-fit assemblies, verifying that cumulative dimensional variation remained within the allowable clearance band defined in the design specification and flagging two features that required tolerance reallocation.

EDUCATION

Master of Science in Aerospace Systems Engineering May 2026

Wright State University, Dayton, OH

GPA: 3.7 | Graduate Research Assistant under CECS Dean Dr. Darryl K. Ahner | Research: eVTOL sizing using MBSE, numerical methods for fluid dynamics

Bachelor of Science in Aeronautical Engineering May 2023

Bharath University, Chennai, TN, India

GPA: 3.5

PROJECT EXPERIENCE

eVTOL Sizing Engine — evtolsizer.live | React, Vite, MATLAB, JavaScript, OpenVSP, CATIA Magic SysML

- Designed and deployed a browser-based eVTOL sizing tool at evtolsizer.live using React/Vite with GitHub Pages hosting, implementing actuator disk hover power, BEM-approximated rotor performance, and ISA atmospheric corrections — the sizing loop converges MTOW within $\pm 5\%$ of three published reference aircraft.
- Built a weight and balance envelope tab computing CG excursion across all passenger and cargo loading permutations, a full metric/imperial unit toggle propagated through all computation modules and chart outputs, and a PDF report generator exporting sizing inputs, computed outputs, and performance charts in a structured review format.

Hoverbike CFD & Structural Analysis | Cradle CFD, HyperMesh, SOLIDWORKS, FEA

- Computed aerodynamic interference between overlapping rotors at three spacing-to-diameter ratios using Cradle CFD, measuring induced velocity loss and rotor-on-rotor thrust degradation relative to an isolated single-rotor baseline.
- Executed FEA in HyperMesh on the duct structural assembly under representative hover and forward-flight loads, evaluated von Mises stress distributions for CFRP and 6061-Al material cases, and identified the critical load path — results published in IJISRT.

Solid Propellant Rocket Performance Study | NASA CEA, MATLAB

- Simulated AP/Al composite propellant combustion using NASA CEA across O/F ratios from 1.0 to 3.0, mapping Isp and flame temperature as functions of oxidizer mass fraction and identifying an optimal O/F range that delivered ~12% Isp gain over the nominal design point.

20-Seater Business Jet — Capstone Design | CATIA V5, OpenVSP, MATLAB

- Executed Class I/II conceptual sizing for a 20-passenger business jet, computing TOGW, fuel fraction, wing area, thrust loading, and aspect ratio, all benchmarked against the Gulfstream G500 as a reference design point.
- Developed empennage sizing and wing loading trade studies, converging on a configuration that met balanced field length, payload-range, and cruise altitude requirements defined in the design specification.

CERTIFICATIONS

- Certified SolidWorks Professional (CSWP) — Dassault Systèmes
- Certified SolidWorks Associate (CSWA) — Dassault Systèmes
- SolidWorks Drawing Tools Professional — Dassault Systèmes
- SolidWorks Additive Manufacturing Associate — Dassault Systèmes
- 3D Swym Associate — Dassault Systèmes
- Python for Data Science, AI & Development — IBM
- Aircraft Performance; Manufacturing Process Technology I & II — NPTEL
- Product Development & Systems Engineering — Udemy
- MATLAB — MathWorks
- American Society of Mechanical Engineers (ASME) — Professional Member